

6-amaliy mashg'ulot. Klassik segmentatsiya

2026–2027 o'quv yili

Muhitni sozlash. Python 3.11 yoki 3.12, faqat CPU. Versiyalar qat'iy belgilangan (requirements.txt): numpy==2.4.4, scipy==1.17.1, pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0, opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
python -m pip install --upgrade pip && pip install -r requirements.txt
```

Mavzu. Segmentatsiyani baholash (IoU, Dice, Hausdorff), Otsu va multi-Otsu, adaptiv ostona (mean, Gauss, Sauvola), integral tasvir, matematik morfologiya, bog'langan komponentlar, masofa transformi va watershed, k-means, SLIC, soha o'stirish.

Maqsadlar. Talaba IoU/Dice ni hisoblaydi va ularning bog'lanishini chiqaradi; Otsu ostonasini gistogrammadan qo'lda topadi; morfologik amallarni va komponentlarni belgilashni kichik niqobda bajaradi; k-means lokal minimumini tushuntiradi; OpenCV/scikit-image'da to'liq klassik zanjirni (ostona → morfologiya → komponentlar → watershed) quradi va natijani haqiqiy niqob bilan sonli baholaydi.

Baholash (100 ball). 1–16-topshiriqlar: har biri 2 ball (32). 17–30-topshiriqlar (kod): 23, 24, 30 (★, amaliy holatlar) — har biri 8 ball (24), qolgan 11 tasi — 4 ball (44). Kod xatosiz ishlashi, seed=42 bilan takrorlanishi va ko'rsatilgan formatda chiqarishi shart.

Kelishuv. $\text{IoU} = \frac{|A \cap G|}{|A \cup G|}$, $\text{Dice} = \frac{2|A \cap G|}{|A| + |G|}$. Otsu: $p_i = n_i/N$, $\omega_0(t) = \sum_{i \leq t} p_i$, $\mu(t) = \sum_{i \leq t} i p_i$, $\sigma_B^2(t) = \omega_0 \omega_1 (\mu_0 - \mu_1)^2 = \frac{(\mu_T \omega_0 - \mu(t))^2}{\omega_0(1 - \omega_0)}$, $\sin f 0: I \leq t$; teng maksimumlarda eng kichik t . Morfologiyada tasvir tashqaris 0. Qutilar (x, y, w, h) ko'rinishida.

A qism. Qo'lda hisoblash, keltirib chiqarish, nazariya

- 1. IoU va Dice.** a) $|A| = 180$, $|G| = 150$, $|A \cap G| = 120$. b) Qutilar $A = (20, 30, 60, 40)$ va $B = (40, 40, 60, 40)$: kesishma, birlashma, IoU. Aniqlash odatda $\text{IoU} \geq 0,5$ da qabul qilinadi — bu juft qabul qilinadimi?
- 2. Bog'lanish.** $\text{Dice} = \frac{2 \text{IoU}}{1 + \text{IoU}}$ ni keltirib chiqaring; IoU = 0,6 uchun Dice, Dice = 0,9 uchun IoU. 256×256 tasvirda obyekt 655 piksel, bashorat bo'sh: piksel aniqligi, IoU, Dice.
- 3. Otsu (8 daraja).** $n = (2, 5, 7, 3, 1, 4, 6, 2)$, $i = 0..7$. μ_T , σ_T^2 ; $t = 0..6$ uchun $\omega_0, \mu_0, \mu_1, \sigma_B^2$ jadvali (4 xona), t^* , σ_W^2 va $\eta = \sigma_B^2 / \sigma_T^2$.
- 4. Dispersiya yoyilishi.** $\sigma_T^2 = \sigma_W^2 + \sigma_B^2$ ni isbotlang va $\omega_0(\mu_0 - \mu_T)^2 + \omega_1(\mu_1 - \mu_T)^2 = \omega_0 \omega_1 (\mu_0 - \mu_1)^2$ ekanini ko'rsating. Nega Otsu barcha ostonalarni $O(L)$ amalda tekshiradi?
- 5. Uch darajali gistogramma.** 20 da 3 piksel, 100 da 5, 200 da 2. Ikki mumkin bo'lgan bo'linish uchun σ_B^2 , σ_T^2 , Otsu ostonasi (qaysi darajalar orasida) va η .
- 6. Integral tasvir.** $I = \begin{pmatrix} 3 & 1 & 4 & 1 \\ 5 & 9 & 2 & 6 \\ 5 & 3 & 5 & 8 \\ 9 & 7 & 9 & 3 \end{pmatrix}$ uchun S . S orqali 1–2-qatorlar va 1–3-ustunlar (0 dan sanab) blokining yig'indisi. ADAPTIVE_THRESH_MEAN_C, 3×3 oyna, $C = 1$: (2, 2) pikselida lokal ostona va THRESH_BINARY natijasi.
- 7. Notekis yoritish.** Siyoh 50, qog'oz 200, yoritish chapdan o'ngga 1,0 dan 0,4 gacha chiziqli. Barcha piksellarni to'g'ri ajratuvchi global t oralig'i (siyoh: $I \leq t$)? Yoritish qaysi qiymatdan kichik bo'lsa bunday t qolmaydi? Adaptiv ostona nima uchun ishlaydi?
- 8. 1D morfologiya.** $f = 11011100111101$, $K = \{-1, 0, 1\}$. Yemirish, kengaytirish, ochish, yopish. Ochish qaysi bo'laklarni o'chirdi va nega?
- 9. 2D ochish.** $A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$, $K = 3 \times 3$ kvadrat. $A \ominus K$, $A \circ K$, pikseller soni. Qaysi piksel(lar) yo'qoldi?
- 10. Bog'langan komponentlar.** $B = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$. 4- va 8-bog'lanishda komponentlar soni va maydonlari; fon komponentlari soni (4- va 8-bog'lanish). Nega obyekt va fon uchun har xil bog'lanish olinadi?
- 11. Ikki o'tishli belgilash.** $\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$, 4-bog'lanish. 1-o'tishdagi vaqtinchalik yorliqlar va ekvivalentliklar; yakuniy komponentlar soni. Bitta o'tishli usul nechta yorliq qoldiradi?
- 12. Masofa transformi.** 5×7 to'rtburchak (fon bilan o'ralgan): L_1 masofa transformi. (0, 0) va (3, 4) nuqtalar orasidagi L_1 , shaxmat (L_∞) va L_2 masofalari. Tegib turgan ikki disk ($r = 10$ va $r = 5$) uchun $D > 0,55 \max D$

markerlari nechta obyektни topadi?

13. **1D watershed.** $D = (1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 3, 3, 2, 1)$. Lokal maksimumlar (markerlar), $f = -D$ relyefidagi ajratish nuqtasi; $D > 0,55 \max D$ global ostonasi nechta marker beradi?
14. **k-means (1D).** $x = (1, 3, 4, 8, 9, 15, 16)$, $K = 2$. a) $c^{(0)} = (1, 3)$ dan Lloyd iteratsiyalari, yakuniy SSE va ostona. b) Barcha 6 boʻlinishni tekshirib global minimumni toping (Otsu mezoni). c) $c^{(0)} = (5, 15)$ dan nima chiqadi? Xulosa.
15. **SLIC.** 480×640 tasvir, $K = 300$: panjara qadami $S = \sqrt{N/K}$, superpiksellar joylashuvi. Masofa $D = \sqrt{d_c^2 + (d_s/S)^2 m^2}$: $d_c = 12$, $d_s = 20$ uchun $m = 10$ va $m = 40$. m oshsa superpiksellar qanday boʻladi? Qidiruv oynasi nega $2S \times 2S$?
16. **Soha oʻstirish va graf kesimi.** $f = (40, 43, 38, 45, 90, 95, 44, 41, 39)$, urugʻ 0-piksel, $|f_i - \bar{f}_R| \leq 8$ (joriy oʻrtacha). Yakuniy soha va $\bar{f}_R$; nega 6–8-piksellar kirmaydi? Graf kesimi vazni $w = \exp(-\Delta I^2 / (2\sigma^2))$, $\sigma = 20$: $\Delta I = 10$ va 40 uchun.

B qism. Dasturlash (Python, OpenCV, scikit-image, scikit-learn)

Umumiy kod (barcha tasvirlar `skimage.data` dan, internet shart emas):

```
from collections import deque
import numpy as np, cv2
from scipy import ndimage as ndi
from scipy.spatial.distance import directed_hausdorff
from skimage import data, filters, segmentation
from skimage.feature import peak_local_max
from sklearn.cluster import KMeans
SEED = 42

def iou(a, b):
    a, b = a > 0, b > 0; return float((a & b).sum() / max((a | b).sum(), 1))
def dice(a, b):
    a, b = a > 0, b > 0; return float(2 * (a & b).sum() / max(a.sum() + b.sum(), 1))
def disks(shape, circles): # circles: [(x, y, r), ...] -> uint8 0/1 niqob
    m = np.zeros(shape, np.uint8)
    for x, y, r in circles: cv2.circle(m, (x, y), r, 1, -1)
    return m
```

Hujjatlar: https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html, https://docs.opencv.org/4.x/d9/d61/tutorial_py_morphological_ops.html, https://docs.opencv.org/4.x/d3/db4/tutorial_py_watershed.html, <https://scikit-image.org/docs/stable/api/skimage.filters.html>, <https://scikit-image.org/docs/stable/api/skimage.segmentation.html>, https://scikit-image.org/docs/stable/auto_examples/applications/plot_coins_segmentation.html, <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>, <https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.label.html>, https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.directed_hausdorff.html.

17. **Otsu qoʻlda.** `otsu_manual(img) -> (t, eta): np.bincount(minlength=256)`, kumulyativ yigʻindilar, σ_B^2 ning $\arg\max_i, \eta = \sigma_B^2(t) / \sigma_T^2$. coins va camera uchun `cv2.threshold(..., THRESH_BINARY+THRESH_OTSU)` va `filters.threshold_otsu` bilan solishtiring; $I > t$ ulushi (4 xona).
18. **Multi-Otsu.** `filters.threshold_multiotsu(camera, classes=3)`, `np.digitize`: ostonalar, sinf ulushlari (4 xona) va oʻrtacha yorqinliklar (1 xona).
19. **Adaptiv ostona.** Sunʼiy hujjat (GT — siyoh niqobi):

```
def make_doc(seed=SEED):
    rng = np.random.default_rng(seed); H, W = 240, 360
    gt = np.zeros((H, W), bool)
    for r in range(15, H - 20, 22):
        c = 10
        while c < W - 20:
            h, w = rng.integers(5, 13), rng.integers(3, 9)
            gt[r:r+h, c:c+w] = True; c += w + int(rng.integers(3, 8))
        v, u = np.mgrid[:H, :W]
        light = 1.0 - 0.5 * (((u - 60) / W)**2 + ((v - 40) / H)**2)
        img = np.clip(np.where(gt, 50.0, 210.0) * light + rng.normal(0, 4, (H, W)), 0, 255).astype(np.uint8)
    return img, gt, float(light.min())
```

Siyoh = qorongʻi: Otsu ($I \leq t$), `adaptiveThreshold(..., MEAN_C / GAUSSIAN_C, THRESH_BINARY_INV, 31, 10)`, Sauvola (`filters.threshold_sauvola(img, window_size=31, k=0.2)`, $I \leq T$). IoU (4 xona); IoU = 1 beradigan global t lar oraligʻi (yoki yoʻqligi). Rasm `out/t19_doc.png`.

20. **Integral tasvir.** `integral_image(img) ((H+1) × (W+1), cumsum, int64)` va `box_mean(S, w)` (chetlarda

kesilgan oyna). `camera: cv2.integral` bilan tengligi, $S[-1, -1]$, $[100:300, 200:400]$ blok yigʻindisi (4 murojaat) va tekshiruvi, $w = 31$ da ichki sohada `ndi.uniform_filter(..., 31, mode='constant')` bilan maksimal farq.

21. **Morfologiya.** `gt = disks((200,300), [(60,60,30),(160,70,25),(240,140,35),(90,150,20)]); rng=default_rng(42);` avval 3% piksel 1 ga (`rng.random(shape) < 0.03`), keyin yangi tasodifiy 3% piksel 0 ga. SE: MORPH_ELLIPSE 5×5 . Xom, ochish, yopish, ochish→yopish, yopish→ochish: IoU, 8-bogʻlangan komponentlar, teshiklar (`binary_fill_holes` farqining komponentlari). Idempotentlik va $A \circ K \subseteq A$ ni tekshiring.
22. **Komponentlarni belgilash.** `two_pass_label(B, conn) -> (labels, n)` (union-find, yoʻl siqish). `B = (default_rng(42).random((64,64)) < 0.45).astype(np.uint8)`; 4 va 8-bogʻlanishda oʻz natijangiz, `ndi.label` va `cv2.connectedComponents`; boʻlinish bir xilmi (yorliqlar juftligi); eng katta komponent; fonning 4-bogʻlangan komponentlari.
23. **★ Tangalarni sanash (watershed).** `data.coins().` a) `GaussianBlur((0,0),1.5)`, `otsu_manual`, `binary_fill_holes`, `ndi.label`: komponentlar va maydoni ≥ 100 boʻlganlar. b) Relyef `filters.sobel(g)`, markerlar: $g < 30 \rightarrow 1$ (fon), $g > 150 \rightarrow 2$ (tanga); `segmentation.watershed`, `binary_fill_holes(ws==2)`, `ndi.label`, maydon ≥ 100 : tangalar soni, maydon min/median/max. Konturlarni `out/t23_coins.png` ga chizing. Nega global Otsu kam sanaydi?
24. **★ Tegib turgan obyektlar.** 220×300 tasvir, disklar (x, y, r) : (60,60,28), (105,60,24), (200,70,16), (230,70,16), (80,160,12), (150,150,30), (205,170,20), (265,160,10) (yorliq i bilan, tartibda chiziladi). Komponentlar, `cv2.distanceTransform(m, DIST_L2, 5)`, global markerlar ($D > 0.55 \max D$), `peak_local_max(D, min_distance=8, labels=m)` markerlari, `watershed(-D, markers, mask=m)` sohalari; har bir disk uchun eng yaxshi IoU (min, oʻrtacha, 3 xona).
25. **Rang boʻyicha k-means.** `data.chelsea() -> Lab (cv2.COLOR_RGB2LAB, float64,  $N \times 3$ ).` `KMeans(n_clusters=K, n_init=4, random_state=42)`, $K = 2..5$: inertia/ N (2 xona), klaster ulushlari (kamayish tartibida, 3 xona). $K = 4$ natijani markaz ranglari bilan `out/t25_kmeans.png` ga saqlang. “Tirsak” qayerda?
26. **SLIC.** `segmentation.slic(astronaut, n_segments=300, compactness=c, start_label=1)`, $c = 1, 10, 40$: superpiksellar soni, oʻrtacha, min va max maydon, ichki Lab standart ogʻishi oʻrtachasi (2 xona), soha boshiga chegara piksellari (`find_boundaries(mode='inner')`). `out/t26_slic.png`.
27. **Baholash oʻlchovlari.** GT: `disks((200,200), [(100,100,40)])`. Bashoratlar: markaz (106,100); radius 46; GT + `disks(..., [(180,20,3)])` dogʻi; boʻsh niqob. IoU, Dice, piksel aniqligi (4 xona) va chegara nuqtalari (`m ^ binary_erosion(m)`) boʻyicha simmetrik Hausdorff (`directed_hausdorff`, 2 xona).
28. **Soha oʻstirish.** `region_grow(img, seed, tau) -> bool` niqob (BFS, 4-bogʻlanish, joriy oʻrtacha bilan). `g = GaussianBlur(camera, (0,0), 1.0)`, urugʻ $(v, u) = (40, 60)$, $\tau = 5, 10, 20$: maydon, ulush, oʻrtacha; `cv2.floodFill(4 | FLOODFILL_FIXED_RANGE | FLOODFILL_MASK_ONLY | (1<<8), loDiff=upDiff=tau)` bilan maydon va IoU.
29. **Otsu va k-means.** 14-topshiriqdagi x uchun Lloyd ($c^{(0)} = (1, 3)$ va $(5, 15)$) va toʻliq qidiruv; `camera` intensivliklarida `KMeans(2, n_init=4, random_state=42)` markazlari oʻrtasini Otsu t bilan solishtiring.
30. **★ Toʻliq zanjir.** `gt = data.binary_blobs(length=256, blob_size_fraction=0.08, volume_fraction=0.25, rng=42)`; fon $60 + 0.25u$, obyekt fon+70, $\mathcal{N}(0, 15^2)$ (`default_rng(42)`), `uint8`. `GaussianBlur((0,0),1.5)`; uch niqob: global Otsu ($> t$); `adaptiveThreshold(GAUSSIAN_C, THRESH_BINARY, 51, -5)`; top-hat (MORPH_ELLIPSE 41×41) + Otsu. Har biri uchun xom IoU, ochish+yopish (5×5 ellips) dan keyingi IoU va 8-bogʻlangan obyektlar soni (GT bilan).